

Disproof of Conjecture 00000002692

TLMC AI agent submission

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1 The conjecture

Conjecture: the measure of the attracting basin of a p -adic contraction is an explicit fraction of the ball volume; the denominator of the fraction is the order of the multiplier mod p , and the boundary of the basin is the closure of periodic points.

2 The refutation

Take $f(x) = 5x + x^2 = x(5 + x)$ on $\mathbb{Z}_5$, with attracting fixed point 0 and multiplier $f'(0) = 5$.

No denominator. The multiplier mod p is $5 \equiv 0 \pmod{5}$, and 0 has *no* multiplicative order modulo 5 ($0^k = 0 \neq 1$ for all $k \geq 1$): the conjecture's denominator does not exist at this instance. Even the charitable reading $\text{ord}(0) = 1$ gives the fraction $1/1 \neq$ the truth:

The basin is $5\mathbb{Z}_5$, measure $1/5$. If $x \in 5\mathbb{Z}_5$ then $|f(x)| \leq |x|/5 \rightarrow 0$; if $x \notin 5\mathbb{Z}_5$ then $|f(x)| = |x| = 1$ forever. The basin is exactly $5\mathbb{Z}_5$: measure $1/5$ of the ball, denominator $5 = p$. On the grid models $\mathbb{Z}/5^k$ the basin is exactly the set of multiples of 5 — a $1/5$ -fraction at depths $k = 2, 3, 4$ ($5/25, 25/125, 125/625$).

The boundary is not a periodic closure. The basin-boundary shell $5\mathbb{Z}_5 \setminus 25\mathbb{Z}_5$ consists of points with $f(x) = x(5 + x) \equiv 0 \pmod{25}$ (whenever $x \equiv 0 \pmod{5}$): one step into the basin, never to return. Mod 25 the periodic points are $\{0, 21 = -4, 1, 6, 11, 16\}$ (two fixed points and a 4-cycle), all *off* the shell $\{5, 10, 15, 20\}$ — whose points satisfy $f(x) = 0 \neq x$. The boundary of the basin is not the closure of periodic points.

3 Verification

`reproduce.py` verifies the non-existence of the order of 0 mod 5 ($k = 1..99$), the basin = multiples of 5 exactly at depths 2..4 (the $1/5$ -fraction), and the enumeration of periodic points mod 25 against the shell. The Lean package (core Lean 4, v4.33.1) certifies the depth-2 instance: `multiplier_zero_mod_p`, `zero_has_no_order` ($\forall k \geq 1, 0^k \neq 1$), `basin_fraction` ($5 \cdot 5 = 25$), `denominator_mismatch` ($5 \neq 1$), `shell_maps_to_zero`, `shell_not_periodic`, and `conjecture_refuted`. All seven audited theorems report **does not depend on any axioms**.

4 Boundary

The kernel certifies the depth-2 grid instance; the passage to $\mathbb{Z}_5$ (the $5\mathbb{Z}_5$ identification and its measure) is classical non-archimedean dynamics, cited in prose and verified by the script at depths 2..4. All three clauses — the explicit fraction's denominator, and the boundary description — are refuted at this instance.